

OCR Further Mathematics
Statistics - Linear Regression
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

- 1(a) At a seaside resort the number X of ice-creams sold and the temperature $Y^{\circ}\text{F}$ were recorded on 20 randomly chosen summer days. The data can be summarised as follows.

$$\Sigma x = 1506$$

$$\Sigma x^2 = 127\,542$$

$$\Sigma y = 1431$$

$$\Sigma y^2 = 140\,451$$

$$\Sigma xy = 111\,297$$

Calculate the equation of the least squares regression line of y on x , giving your answer in the form $y = a + bx$. [3]

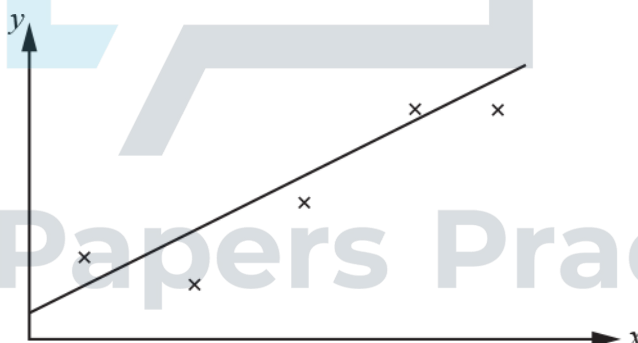
- (b) Explain the significance for the regression line of the quantity $\Sigma[y_i - (ax_i + b)]^2$. [1]

- (c) It is decided to measure the temperature in degrees Centigrade instead of degrees Fahrenheit. If the same temperature is measured both as f° Fahrenheit and c° Centigrade, the relationship between f and c is

$$c = \frac{5}{9}(f - 32)$$

Find the equation of the new regression line. [2]

- 2(a) Using the scatter diagram below, explain what is meant by least squares in the context of a regression line of y on x .



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[2]

(b) A set of bivariate data (t, u) is summarised as follows.

$$\begin{array}{lll}
 n = 5 & \Sigma t = 35 & \Sigma u = 54 \\
 \Sigma t^2 = 285 & \Sigma u^2 = 758 & \Sigma tu = 460
 \end{array}$$

(i) Calculate the equation of the regression line of u on t .

[3]

(ii) The variables t and u are now scaled using the following scaling.

$$v = 2t, w = u + 4$$

Find the equation of the regression line of w on v , giving your equation in the form $w = f(v)$.

[2]

3(a) An investor obtains data about the profits of 8 randomly chosen investment accounts over two one-year periods.

The profit in the first year for each account is $p\%$ and the profit in the second year for each account is $q\%$.

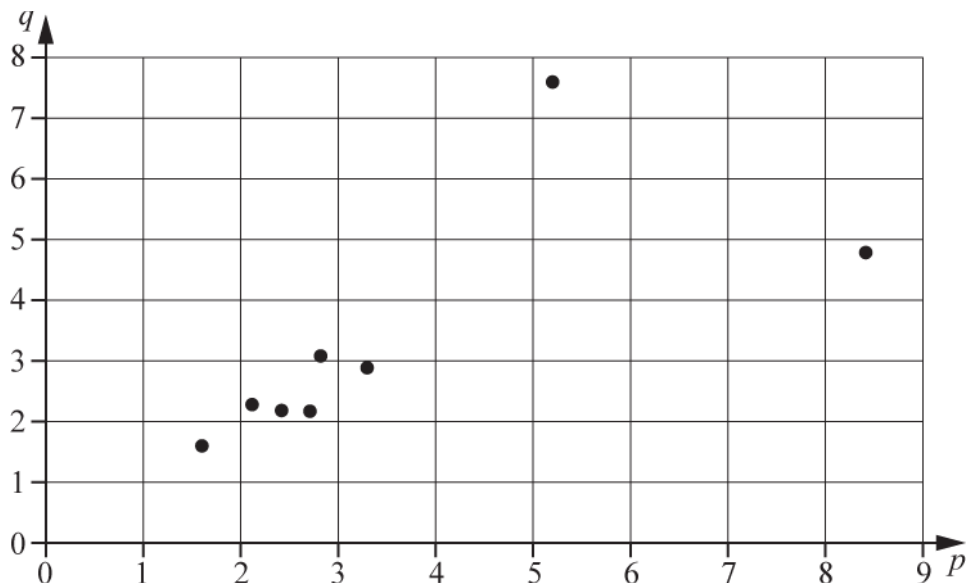
The results are shown in the table and in the scatter diagram.

Account	A	B	C	D	E	F	G	H
p	1.6	2.1	2.4	2.7	2.8	3.3	5.2	8.4
q	1.6	2.3	2.2	2.2	3.1	2.9	7.6	4.8

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$$\begin{array}{llllll}
 n = 8 & \Sigma p = 28.5 & \Sigma q = 26.7 & \Sigma p^2 = 136.35 & \Sigma q^2 = 116.35 & \Sigma pq = 116.70
 \end{array}$$



State which, if either, of the variables p and q is independent.

[1]

(b) Calculate the equation of the regression line of q on p .

[3]

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(c) (i) Use the regression line to estimate the value of q for an investment account for which $p = 2.5$. [1]

(ii) Give two reasons why this estimate could be considered reliable. [2]

(d) Comment on the reliability of using the regression line to predict the value of q when $p = 7.0$. [2]

4(a) A set of bivariate data (X , Y) is summarised as follows.

$$n = 25, \sum x = 9.975, \sum y = 11.175, \sum x^2 = 5.725, \sum y^2 = 46.200, \sum xy = 11.575$$

Calculate the value of Pearson's product-moment correlation coefficient. [1]

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(b) Calculate the equation of the regression line of y on x . [2]

- (c) It is desired to know whether the regression line of y on x will provide a reliable estimate of y when $x = 0.75$.

State one reason for believing that the estimate will be reliable.

[1]

- (d) State what further information is needed in order to determine whether the estimate is reliable.

[1]

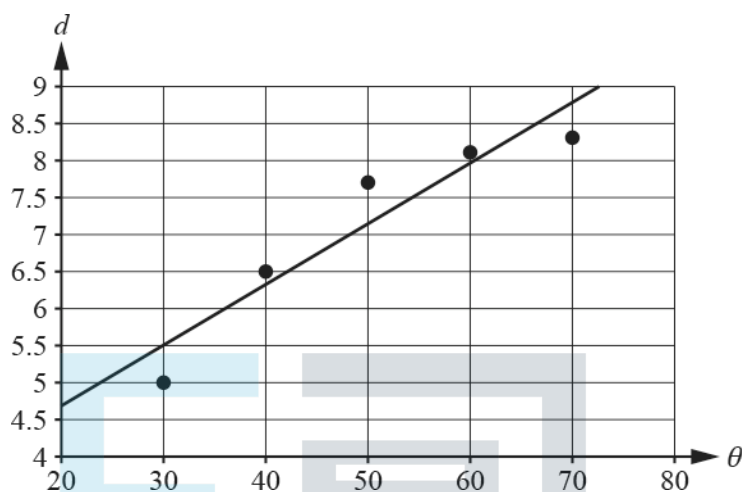


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- 5 An experiment involves releasing a coin on a sloping plane so that it slides down the slope and then slides along a horizontal plane at the bottom of the slope before coming to rest. The angle θ° of the sloping plane is varied, and for each value of θ , the distance d cm the coin slides on the horizontal plane is recorded. A scatter diagram to illustrate the results of the experiment is shown below, together with the least squares regression line of d on θ .



- (a) State which two of the following correctly describe the variable θ .

Controlled variable

Correlation coefficient

Dependent variable

Independent variable

Response variable

Regression coefficient

[1]

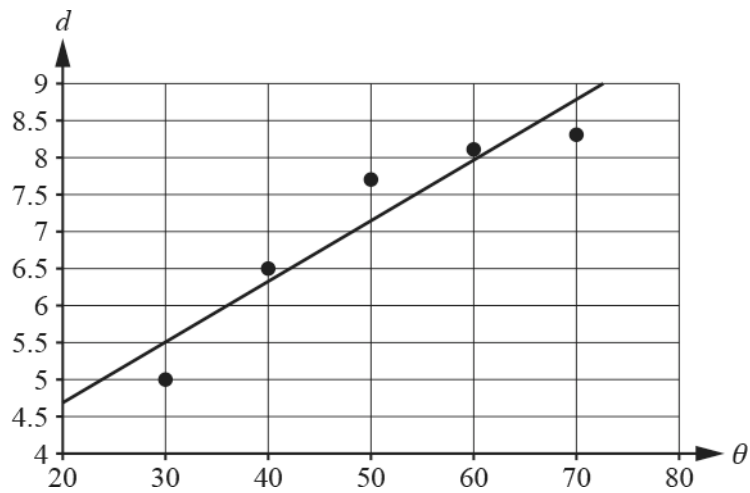
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The least squares regression line of d on θ has equation $d = 1.96 + 0.11\theta$.

(b) Use the diagram to explain the term “least squares”.

[2]



(c) State what difference, if any, it would make to the equation of the regression line if d were measured in inches rather than centimetres. (1 inch $\approx$ 2.54 cm).

[1]

- 6 An environmentalist measures the mean concentration, c milligrams per litre, of a particular chemical in a group of rivers, and the mean mass, m pounds, of fish of a certain species found in those rivers. The results are given in the table.

c	1.94	1.78	1.62	1.51	1.52	1.4
m	6.5	7.2	7.4	7.6	8.3	9.7

- (a) State which, if either, of m and c is an independent variable.

[1]

- (b) Calculate the equation of the least squares regression line of c on m .

[3]

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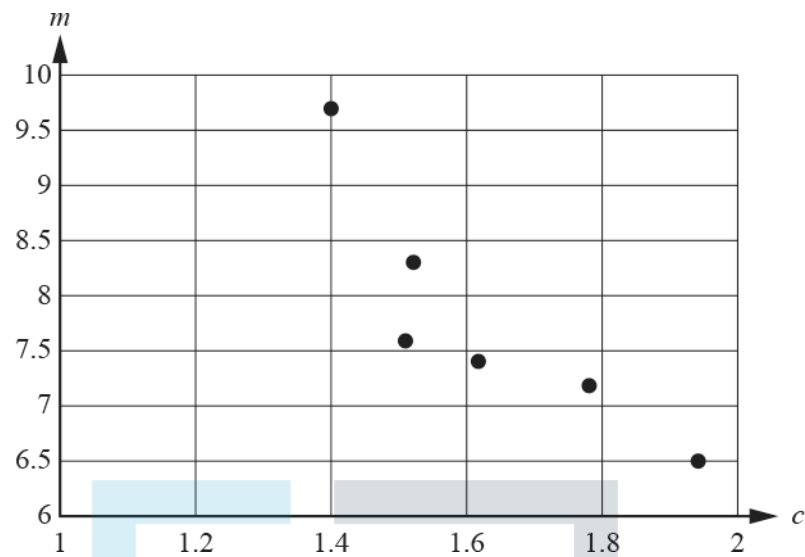
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- (c) State what effect, if any, there would be on your answer to part (b) if the masses of the fish had been recorded in kilograms rather than pounds. (1 kg $\approx$ 2.2 pounds.)

[1]

- (d) The data is illustrated in the scatter diagram. Explain what is meant by 'least squares', illustrating your answer using the diagram below.

[3]



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- 7 The values of a set of bivariate data (x_i, y_i) can be summarised by

$$n = 50, \sum x = 1270, \sum y = 5173, \sum x^2 = 42767, \sum y^2 = 701301, \sum xy = 173161.$$

Ten independent observations of Y are obtained, all corresponding to $x = 20$. It may be assumed that the variance of Y is 1.9, independently of the value of x . Find a 95% confidence interval for the mean $\bar{Y}$ of the 10 observations of Y .

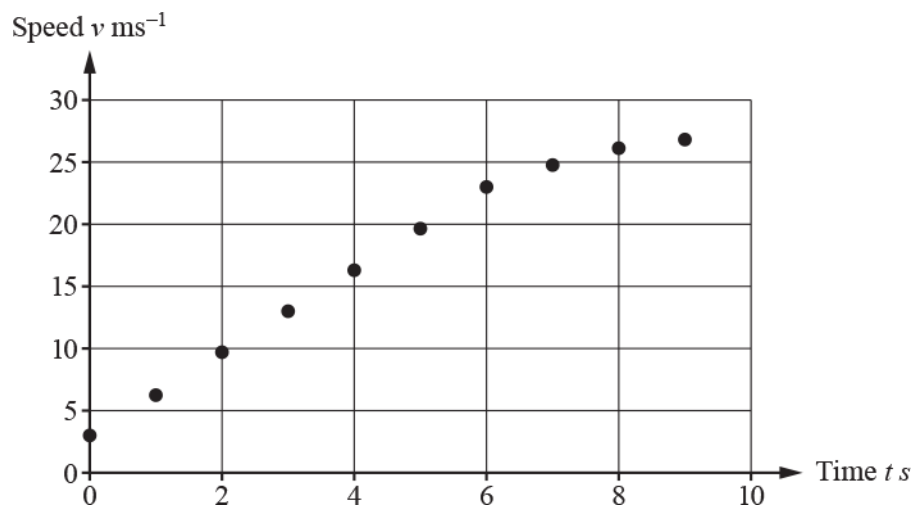
[8]

- 8 The speed $v \text{ ms}^{-1}$ of a car at time t seconds after it starts to accelerate was measured at 1-second intervals. The results are shown in the following diagram.

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(a) State whether t or v or neither is a controlled variable.

[1]

The value of the product moment correlation coefficient r for the data is 0.987 correct to 3 significant figures.

(b) The speed of the car is converted to miles per hour and the time to minutes. State the value of r for the converted data.

[1]

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(c) State the value of Spearman's rank correlation coefficient r_s for the data.

[1]

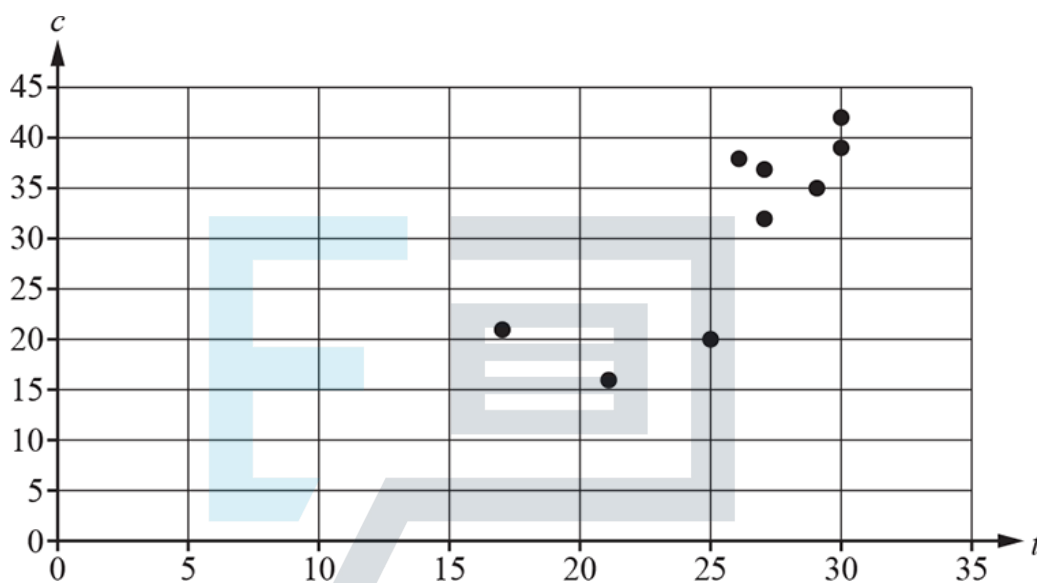
(d) What information does r give about the data that is not given by r_s ?

[1]

- 9 A shop manager recorded the maximum daytime temperature T °C and the number C of ice creams sold on 9 summer days.

The results are given in the table and illustrated in the scatter diagram.

T	17	21	25	26	27	27	29	30	30
C	21	16	20	38	32	37	35	39	42



$$n = 9, \Sigma t = 232, \Sigma c = 280, \Sigma t^2 = 6130, \Sigma c^2 = 9444, \Sigma tc = 7489$$

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- (a) State, with a reason, whether one of the variables C or T is likely to be dependent upon the other. [1]

- (b) Calculate Pearson's product-moment correlation coefficient r for the data. [2]

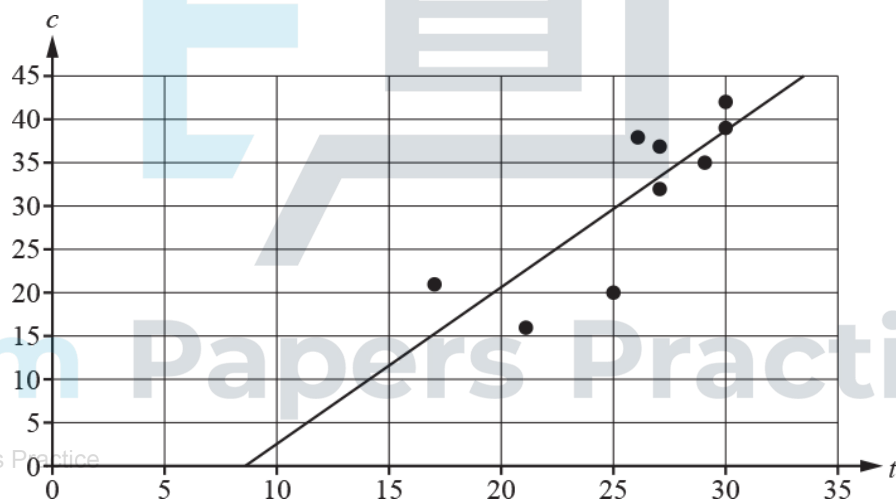
- (c) State with a reason what the value of r would have been if the temperature had been measured in °F rather than °C. [1]

(d) Calculate the equation of the least squares regression line of c on t .

[2]

(e) The regression line is drawn on the copy of the scatter diagram below. Use this diagram to explain what is meant by “least squares”.

[2]



10 The table below shows the typical stopping distances d metres for a particular car travelling at v miles per hour.

v	20	30	40	50	60	70
d	13	24	36	52	72	94

(a) State each of the following words that describe the variable v .

Independent

Dependent

Controlled

Response

[1]

(b) Calculate the equation of the regression line of d on v .

[2]

(c) Use the equation found in part (b) to estimate the typical stopping distance when this car is travelling at 45 miles per hour.

[1]

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It is given that the product moment correlation coefficient for the data is 0.990 correct to three significant figures.

(d) Explain whether your estimate found in part (c) is reliable.

[2]

- 11 The following table gives the mean per capita consumption of mozzarella cheese per annum, x pounds, and the number of civil engineering doctorates awarded, y , in the United States in each of 10 years.

x	9.3	9.7	9.7	9.7	9.9	10.2	10.5	11.0	10.6	10.6
y	480	501	540	552	547	622	655	701	712	708

source: www.tylervigen.com

- (a) Find the equation of the regression line of y on x .

[2]

You are given that the product moment correlation coefficient is 0.959.

- (b) Explain whether this value would be different if x is measured in kilograms instead of pounds.

[1]

It is desired to carry out a hypothesis test to investigate whether there is correlation between these two variables.

- (c) Assume that the data is a random sample of all years.

- (i) Carry out the test at the 10% significance level.

[6]

- (ii) Explain whether your conclusion suggests that manufacturers of mozzarella cheese could increase consumption by sponsoring doctoral candidates in civil engineering. [1]

- 12 The table shows some of the values of the seasonally adjusted Unemployment Rate (UR), x %, and the Consumer Price Index (CPI), y %, in the United Kingdom from April 2008 to July 2010.

Date	April 2008	July 2008	October 2008	January 2009	April 2009	July 2009	October 2009	January 2010	April 2010	July 2010
UR, x %	5.2	5.7	6.1	6.8	7.5	7.8	7.8	7.9	7.8	7.7
CPI, y %	3.0	4.4	4.5	3.0	2.3	1.8	1.5	3.5	3.7	3.1

These data are summarised below.

$$n = 10 \quad \Sigma x = 70.3 \quad \Sigma x^2 = 503.45 \quad \Sigma y = 30.8 \quad \Sigma y^2 = 103.94 \quad \Sigma xy = 211.9$$

- (i) Calculate the product moment correlation coefficient, r , for the data, showing that $-0.6 < r < -0.5$.

[3]

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- (ii) Karen says “The negative value of r shows that when the Unemployment Rate increases, it causes the Consumer Price Index to decrease.” Give a criticism of this statement.

[1]

(iii) (a) Calculate the equation of the regression line of x on y . [3]

(b) Use your equation to estimate the value of the Unemployment Rate in a month when the Consumer Price Index is 4.0%.

[2]



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- 13 The Gross Domestic Product per Capita (GDP), x dollars, and the Infant Mortality Rate per thousand (IMR), y , of 6 African countries were recorded and summarised as follows.

$$n = 6$$

$$\Sigma x = 7000$$

$$\Sigma x^2 = 8\,700\,000$$

$$\Sigma y = 456$$

$$\Sigma y^2 = 36\,262$$

$$\Sigma xy = 509\,900$$

- (i) Calculate the equation of the regression line of y on x for these 6 countries.

[4]

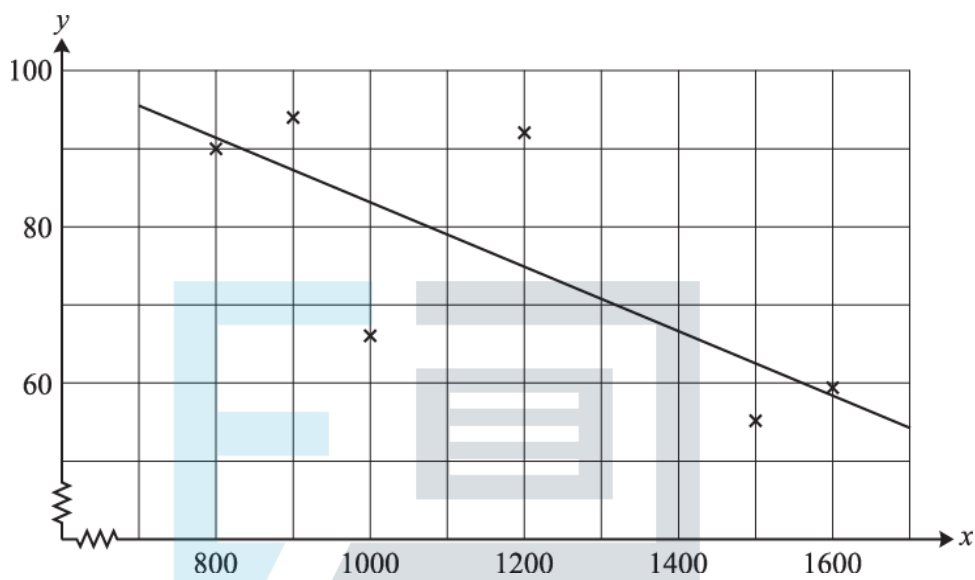


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The original data were plotted on a scatter diagram and the regression line of y on x was drawn, as shown below.



- (ii) The GDP for another country, Tanzania, is 1300 dollars. Use the regression line in the diagram to estimate the IMR of Tanzania.

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[1]

- (iii) The GDP for Nigeria is 2400 dollars. Give two reasons why the regression line is unlikely to give a reliable estimate for the IMR for Nigeria.

[2]

- (iv) The actual value of the IMR for Tanzania is 96. The data for Tanzania ($x = 1300$, $y = 96$) is now included with the original 6 countries. Calculate the value of the product moment correlation coefficient, r , for all 7 countries.

[4]

- (v) The IMR is now redefined as the infant mortality rate per hundred instead of per thousand, and the value of r is recalculated for all 7 countries. Without calculation state what effect, if any, this would have on the value of r found in part (iv).

[1]

14

- (i) The table shows the amount, x , in hundreds of pounds, spent on heating and the number of absences, y , at a factory during each month in 2014.

Amount, x , spent on heating (£ hundreds)	21	23	19	15	14	5	2	10	9	20	18	23
Number of absences, y	23	25	18	18	12	10	4	9	11	15	20	26

$$n = 12$$

$$\sum x = 179$$

$$\sum x^2 = 3215$$

$$\sum y = 191$$

$$\sum y^2 = 3565$$

$$\sum xy = 3343$$

- (a) Calculate r , the product moment correlation coefficient, showing that $r > 0.92$.

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[3]

- (b) A manager says, 'The value of r shows that spending more money on heating causes more absences, so we should spend less on heating.' Comment on this claim.

[1]

- (ii) The months in 2014 were numbered 1, 2, 3, ..., 12. The output, z , in suitable units was recorded along with the month number, n , for each month in 2014. The equation of the regression line of z on n was found to be $z = 0.6n + 17$.

- (a) Use this equation to explain whether output generally increased or decreased over these months.

[1]

- (b) Find the mean of n and use the equation of the regression line to calculate the mean of z .

[3]

- (c) Hence calculate the total output in 2014.

[2]

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- 15 For the top 6 clubs in the 2010/11 season of the English Premier League, the table shows the annual salary, £x million, of the highest paid player and the number of points scored, y.

Club	Manchester United	Manchester City	Chelsea	Arsenal	Tottenham	Liverpool
x	5.6	7.4	6.5	4.1	3.6	6.5
y	80	71	71	68	62	58

$$n = 6 \qquad \Sigma x = 33.7 \qquad \Sigma x^2 = 200.39 \qquad \Sigma y = 410 \qquad \Sigma y^2 = 28\,314 \qquad \Sigma xy = 2313.9$$

- (i) Use a suitable formula to calculate the product moment correlation coefficient, r , between x and y , showing that $0 < r < 0.2$.

[3]

- (ii) State what this value of r shows in this context.

[1]

- (iii) A fan suggests that the data should be used to draw a regression line in order to estimate the number of points that would be scored by another Premier League club, whose highest paid player's salary is £1.7 million. Give two reasons why such an estimate would be unlikely to be reliable.

[2]

- 16 The table shows the load a lorry was carrying, x tonnes, and the fuel economy, y km per litre, for 8 different journeys. You should assume that neither variable is controlled.

Load (x tonnes)	5.1	5.8	6.5	7.1	7.6	8.4	9.5	10.5
Fuel economy (y km per litre)	6.2	6.1	5.9	5.6	5.3	5.4	5.3	5.1

$$n = 8 \qquad \Sigma x = 60.5 \qquad \Sigma y = 44.9 \qquad \Sigma x^2 = 481.13 \qquad \Sigma y^2 = 253.17 \qquad \Sigma xy = 334.65$$

- (i) Calculate the equation of the regression line of y on x .

[4]

- (ii) Estimate the fuel economy for a load of 9.2 tonnes.

[2]

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- (iii) An analyst calculated the equation of the regression line of x on y . Without calculating this equation, state the coordinates of the point where the two regression lines intersect.

[1]

- (iv) Describe briefly the method required to estimate the load when the fuel economy is 5.8 km per litre.

[2]

- 17 Tariq collected information about typical prices, £ y million, of four-bedroomed houses at varying distances, x miles, from a large city. He chose houses at 10-mile intervals from the city. His results are shown below.

x	10	20	30	40	50	60	70	80
y	1.2	1.4	1.2	0.9	0.8	0.5	0.5	0.3

$$n = 8$$

$$\Sigma x = 360$$

$$\Sigma x^2 = 20\,400$$

$$\Sigma y = 6.8$$

$$\Sigma y^2 = 6.88$$

$$\Sigma xy = 241$$

- (i) Use an appropriate formula to calculate the product moment correlation coefficient, r , showing that $-1.0 < r < -0.9$.

[3]

- (ii) State what this value of r shows in this context.

[1]

- (iii) Tariq decides to recalculate the value of r with the house prices measured in hundreds of thousands of pounds, instead of millions of pounds. State what effect, if any, this will have on the value of r .

[1]

- (iv) Calculate the equation of the regression line of y on x .

[3]

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- (v) Explain why the regression line of y on x , rather than x on y , should be used for estimating a value of x from a given value of y .

[1]

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